Year 7
Mathematics

Unit 3: Equations, Functions and Formulae

3.3 Using Formulae

Using Formulae

STARTER · 5 MIN

Date: _____ **Title: Using Formulae**

LEARNING OBJECTIVES

- To be able to substitute numbers into formulae

STARTER *Answer in your book. No calculators.*

1. Work out 4×7
2. Work out $2 \times 5 + 3$
3. Work out $3 \times 6 - 2$
4. Work out $4^2 + 1$

Starter — Answers

STARTER

Tick or correct in green pen

1. **28**
2. **13**
3. **16**
4. **17**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute a number into a one-step formula	☐
SILVER	I can substitute into formulae with two letters or powers	☐
GOLD	I can substitute negative numbers into formulae	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Using Formulae

INTRODUCTION · 3 MIN

A formula is a rule linking quantities. Substitute: replace each letter with its number.

Formula

A rule with an = sign.

$P = 4s$

Substitute

Put the number in place of the letter.

$s = 5$: $P = 4 \times 5 = 20$

Hidden ×

$3x$ means $3 \times x$.

$x = 2$: $3x = 3 \times 2 = 6$

Order

Powers, then × and ÷, then + and −.

$x = 3$: $x^2 + 1 = 9 + 1 = 10$

Examples 1 and 2 — Substituting into Formulae

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint 4*s* means 4 × *s* × before + and −

Example 1

$P = 4s$

Find P when $s = 5$.

Example 2

$C = 2n + 3$

Find C when $n = 4$.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Replace *s* with 5

$P = 4 \times 5$

$= \mathbf{20}$

Tip: $4s$ means $4 \times s$.

Example 2

Multiply first, then add

$C = 2 \times 4 + 3$

$= 8 + 3$

$= \mathbf{11}$

Tip: write the substitution out.

Task 1

PRACTICE · 9 MIN

BRONZE

Substitute to find the value

Hint $4s$ means $4 \times s$ × before + and −

1. $P = 4s,\ s = 7$
2. $C = 2n + 3,\ n = 6$
3. $y = 3x - 2,\ x = 4$
4. $A = l \times w,\ l = 6,\ w = 3$
5. $T = 5m + 1,\ m = 3$

★ $y = 3x - 2,\ x = 10$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $P = 28$
2. $C = 15$
3. $y = 10$
4. $A = 18$
5. $T = 16$

★ $y = 28$

Examples 3 and 4 — Two Letters and Powers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Work out each term $3^2 = 3 \times 3$

Example 3

$F = 2a + 3b$

Find F when $a = 4$ and $b = 5$.

Example 4

$y = x^2 + 1$

Find y when $x = 3$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Substitute both letters

$F = 2 \times 4 + 3 \times 5$

$= 8 + 15$

$\mathbf{= 23}$

Tip: work out each term, then add.

Example 4

Square first, then add

$y = 3^2 + 1$

$= 9 + 1$

$\mathbf{= 10}$

Tip: $3^2 = 3 \times 3$, not 3×2.

Task 2

PRACTICE · 9 MIN

SILVER

Substitute to find the value

Hint Work out each term $3^2 = 3 \times 3$

1. $F = 2a + 3b,\ a = 1,\ b = 6$
2. $y = x^2 + 1,\ x = 5$
3. $W = 4p - q,\ p = 3,\ q = 2$
4. $A = l \times w,\ l = 8,\ w = 2.5$
5. $y = 2x^2,\ x = 3$

★ $D = s \times t,\ s = 60,\ t = 3$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $F = 20$
2. $y = 26$
3. $W = 10$
4. $A = 20$
5. $y = 18$

★ $D = 180$

Examples 5 and 6 — Negative Numbers in Formulae

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Use a bracket: $(-3)^2$ Negative squared: positive

Example 5

(2 marks)

$y = x^2 + 4$

Find y when $x = -3$.

Example 6

(2 marks)

$E = a^2 + 2b$

Find E when $a = -3$ and $b = 4$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Use a bracket for the negative

$$y = (-3)^2 + 4$$
$$= 9 + 4$$
$$\mathbf{= 13}$$

Tip: $(-3)^2 = -3 \times -3 = 9$.

Example 6

Bracket, square, then add

$$E = (-3)^2 + 2 \times 4$$
$$= 9 + 8$$
$$\mathbf{= 17}$$

Tip: a negative squared is positive.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: substitute*

Hint Use a bracket: $(-3)^2$ Negative squared: positive

1. $y = x^2 + 4,\ x = -2$

2. $y = 3x + 10,\ x = -4$

3. $T = a - b,\ a = 3,\ b = -5$

4. $E = a^2 + 2b,\ a = -1,\ b = 3$

5. $P = 2l + 2w,\ l = 7,\ w = 4.5$

★ $M = \dfrac{a + b}{2},\ a = 9,\ b = -3$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $y = 8$
2. $y = -2$
3. $T = 8$
4. $E = 7$
5. $P = 23$
- ★ $M = 3$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $P = 4s$. Find P when $s = 9$.

SILVER $C = 2n + 3$. Find C when $n = 10$.

GOLD $y = x^2 + 4$. Find y when $x = -3$.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

$P = 36$

SILVER

$C = 2 \times 10 + 3$

$= 23$

GOLD

$y = (-3)^2 + 4$

$= 13$

Exam Practice — Substitution

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint 4*s* means 4 × *s* × before + and −

Question 1

(1 mark)

The perimeter of a square is $P = 4s$.

Work out P when $s = 6$.

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Question 2

(2 marks)

$y = 3x - 2$

Work out the value of y when $x = 5$.

.....

Exam Practice — Formulae

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Work out each term $3^2 = 3 \times 3$

Question 3

(2 marks)

$F = 2a + 3b$

Work out F when $a = 6$ and $b = 4$.

.....

Question 4

(2 marks)

The area of a rectangle is $A = l \times w$.

Work out A when $l = 12$ and $w = 3.5$

.....

Exam Practice — Negative Values

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Use a bracket: $(-3)^2$
Negative squared: positive

Question 5

(2 marks)

$E = a^2 + 2b$

Work out E when $a = -3$ and $b = 4$.

.....

Question 6

(3 marks)

$y = x^2 + 4$

(a) Work out y when $x = -2$. (1)

.....

(b) Tom gets -5 for $x = -3$. Why? (2)

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Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$P = 4 \times 6 = 24$ **B1**

Tip: 4*s* means 4 × *s*.

Question 2

$y = 3 \times 5 - 2$ **M1**

$= 13$ **A1**

Tip: multiply before you subtract.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$F = 2 \times 6 + 3 \times 4$ **M1**

$\mathbf{= 24}$ **A1**

Tip: 12 + 12 = 24.

Question 4

$A = 12 \times 3.5$ **M1**

$\mathbf{= 42}$ **A1**

Tip: 12 × 3.5 = 36 + 6.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$E = (-3)^2 + 2 \times 4$ — M1

$= 9 + 8 = 17$ — A1

Tip: $(-3)^2 = 9$.

Question 6

(a) $y = 4 + 4 = 8$ — B1

(b) He did $-3^2 = -9$ — B1

$(-3)^2 = 9$, so $y = 13$ — B1

Tip: always put a negative in a bracket.